or that spin and implode simultaneously. Although these objects do not exist in the real Universe, my calculations about them, done in fits and starts, gradually brought understanding.

By 1972 the truth was evident: Only if an object is compressed in *all three* of its spatial directions, north–south, east–west, and up–down (for example, if it is compressed spherically), can gravity become so strong that it overwhelms all forms of internal pressure. If, instead, the object is compressed in only two spatial directions (for example, if it is compressed cylindrically into a long thin thread), gravity grows strong, but not nearly strong enough to win the battle with pressure. Very modest pressure, whether due to hot gas, electron degeneracy, or magnetic field lines, can easily overwhelm gravity and make the cylindrical object explode. And if the object is compressed in only a single direction, into a very thin pancake, pressure will overwhelm gravity even more easily.

My calculations showed this clearly and unequivocally in the case of spheres, infinitely long cylinders, and infinitely extended pancakes. For such objects, the calculations were manageable. Much harder to compute—indeed, far beyond my talents—were nonspherical objects of finite size. But physical intuition emerging from my calculations and from calculations by my youthful comrades told me what to expect. That expectation I formulated as a *hoop conjecture:*

Take any kind of object you might wish—a star, a cluster of stars, a bundle of magnetic field lines, or whatever. Measure the object's mass, for example, by measuring the strength of its gravitational pull on orbiting planets. Compute from that mass the object's critical circumference (18.5 kilometers times the object's mass in units of the mass of the Sun). If the object were spherical (which it is not) and were to implode or be squeezed, it would form a black hole when it gets compressed inside this critical circumference. What happens if the object is not spherical? The hoop conjecture purports to give an answer (Figure 7.2).

Construct a hoop with circumference equal to the critical circumference of your object. Then try to place the object at the center of the hoop, and try to rotate the hoop completely around the object. If you succeed, then the object must already have created a black-hole horizon around itself. If you fail, then the object is not yet compact enough to create a black hole.

In other words, the hoop conjecture claims that, if an object (a star, a star cluster, or whatever) gets compressed in a highly nonspherical manner,